

3. (a) The table shows the names, molecular formulae and structural formulae of some alkanes and alkenes.

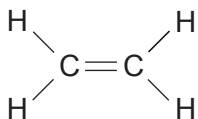
Name	Molecular formula	Structural formula
methane	CH_4	<pre> H H — C — H H </pre>
.....	C_2H_6	<pre> H H H — C — C — H H H </pre>
propene	C_3H_6	
butene		<pre> H H H H \ / C = C — C — C — H / \ H H H H </pre>

Complete the table by filling in the missing name, molecular formula and structural formula.

[3]



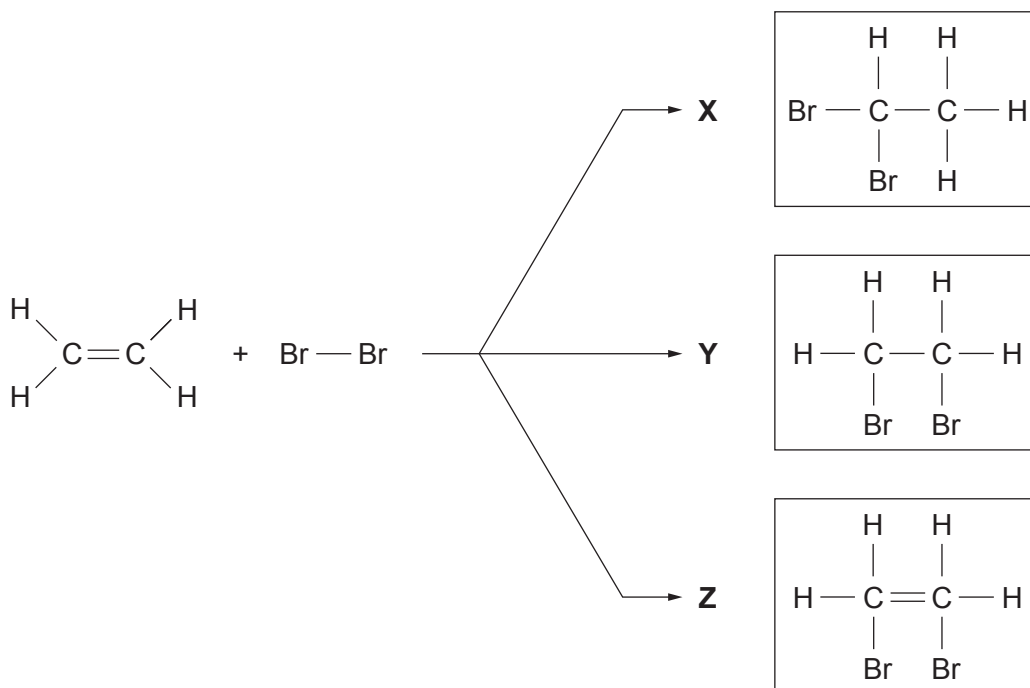
(b) The structure of ethene, C_2H_4 , is shown below.



(i) Underline the correct word in brackets to complete the sentence. [1]

When orange bromine water is added to ethene, the solution becomes
(**darker** / **milky** / **colourless**).

(ii) I. Give the letter, **X**, **Y** or **Z**, that shows the product formed when bromine reacts with ethene. [1]



Letter

II. Give the reason why the product of the reaction between bromine and ethene is **not** a hydrocarbon. [1]

.....
.....



- (c) The electronic structures of carbon and hydrogen are shown in the table.

Element	Electronic structure
carbon	2,4
hydrogen	1

Use this information to complete the diagram to show how the atoms bond in a molecule of methane, CH₄. [1]

